

Break-Even Analysis

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2026-02-26

Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Numerical Methods

Universe

- See problem description

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

A small business is considering launching a new product. Fixed costs are \$25,000 (tooling, marketing). Variable cost per unit is \$12. Selling price per unit is \$35. What is the break-even quantity? How does the break-even point shift if the price drops 10% or variable costs increase 20%?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Break-even analysis	<i>finding the production volume where profit is zero</i>
(2) Symbolic computation	<i>deriving exact formulas with SymPy</i>
(3) Sensitivity analysis	<i>understanding how parameter changes affect the break-even point</i>
(4) Contribution margin	<i>Price minus Variable cost per unit</i>
(5) Business decision modeling	<i>translating financial questions into mathematical models</i>

Approach

Suggested approachSymbolic (SymPy) + Numerical (NumPy) Break-Even Analysis

Complexity class:

Available tools: Symbolic

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	Yes
Algorithm:	Symbolic (SymPy) + Numerical (NumPy) Break-Even Analysis
Time:	0.1147s

Solution Details

Method: Symbolic algebra via SymPy to derive the closed-form break-even formula, then numerical evaluation with NumPy for sensitivity analysis. Break-even occurs where $\text{Revenue}(Q) = \text{Cost}(Q)$, i.e., $P \cdot Q = F + V \cdot Q$, yielding $Q^* = F / (P - V)$. Sensitivity sweeps over price and variable cost perturbations.

Verification

✓ Feasibility All constraints satisfied

3. Interpretation

The Question

A small business is considering launching a new product.

The Answer

A feasible solution was found.

What This Means

- Exact symbolic break-even quantity: $Q = \text{Fixed} / (\text{Price} - \text{Variable})$
- Base case: ~1087 units
- Sensitivity table showing break-even under price drops and cost increases
- Profit projections at various production volumes

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Sensitivity Analysis

Parameter	Current	Change	New Obj.	Class
worst_case	present	removed		robust

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution